

Hence, the class of inhomogeneous Calderón–Zygmund singular integral operators is smaller than the class of Calderón–Zygmund singular integral operators. This kind of operator includes some of the pseudo-differential operators, see [3, 7]. Then, if  $T$  is an inhomogeneous Calderón–Zygmund singular integral operator, the authors in [3, 4] found necessary and sufficient conditions for the boundedness of such operators on  $h^p(\mathbb{R}^n)$  for all  $0 < p \leq 1$ , improving and extending the work [7].

The core of proving the boundedness of a Calderón–Zygmund singular integral operator  $T$  on Hardy spaces or local Hardy spaces is to show that  $T$  maps atoms to molecules. This strategy works perfectly in  $H^p(\mathbb{R}^n)$ , under the assumption that the operator satisfies  $T^*(x^\alpha) = 0$ , i.e.,

$$\int_{\mathbb{R}^n} Ta(x)x^\alpha dx = 0, \quad \forall f \in L_{N_p}^2(\mathbb{R}^n) \text{ and } |\alpha| \leq N_p.$$

This condition is actually necessary and sufficient for the boundedness of such operators in  $H^p(\mathbb{R}^n)$ . However, a local version of such cancellation condition on  $T$  is too strong in the local setting (for instance, for pseudodifferential operators). The key to find less restrictive necessary and sufficient conditions for the boundedness of inhomogeneous Calderón–Zygmund operators in the local setting, was to consider approximate atoms and molecules. To do this, if  $a$  is an atom in  $h^p(\mathbb{R}^n)$ , we show that only the conditions imposed on the kernel imply that  $Ta$  satisfy the size conditions of the molecule. Then, the additional cancellation condition that  $T^*((\cdot - c)^\alpha)$  satisfy a local Campanato-type estimate will show the approximate moment condition of the molecule (see [3, Section 5]). On the other hand, if we suppose that  $Ta$  satisfies the size condition of molecules, then  $T^*((\cdot - c)^\alpha)$  must satisfy the local Campanato type conditions, see [4, Theorem 2]. In a more general setting, the authors in [4, Theorem 1] also proved the necessity of this local Campanato condition for general linear and bounded operators in  $h^p(\mathbb{R}^n)$  that maps atoms into *pseudo-molecules*, a generalization of *pre-molecules*, that will be defined in Definition 3.

Motivated by these results, the goal of this paper is to further investigate this relation between the the size conditions of the kernel and the property of the operator to map atoms into molecules. This work is inspired by a related question answered in the setting of Triebel-Lizorkin spaces  $\dot{F}_p^{\alpha,q}(\mathbb{R}^n)$  for  $\alpha \in (0, 1)$  and  $1 < p, q < \infty$  in [9]. In particular, the authors showed in Theorem 1.16 that a linear operator  $T$  which is continuous from  $C_c^\infty(\mathbb{R}^n)$  with mean 0 to the distribution spaces mapping certain smooth atoms to smooth molecules, then the Schwartz kernel of the restriction on  $C_c^\infty(\mathbb{R}^n)$  of the extension of  $T$  on  $\dot{F}_p^{\alpha,q}(\mathbb{R}^n)$  indeed satisfies the classical Calderón–Zygmund estimates  $|K(x, y)| \leq C|x - y|^{-n}$  and  $|K(x, y) - K(x, z)| \leq C|y - z|^\delta|x - y|^{-n-\delta}$  whenever  $2|y - z| \leq |x - y|$ . However, the necessary conditions on the kernel of an operator to map atoms to molecules for (local) Hardy spaces are unclear.

To motivate our main result, we start by showing that under a suitable integral-type condition on the kernel and some cancellation condition on  $T$ , then  $T$  maps atoms to approximate molecules.

**Theorem 1.** *Let  $0 < p \leq 1 \leq q \leq 2$ ,  $1 \leq s \leq 2$  with  $p \neq s$ , and  $T$  be a Calderón–Zygmund singular integral operator. If the kernel of  $T$  satisfies*

- (i) *there exist a collection of polynomials (of degree at most  $N_p$ )  $\{P_{K,B}(x, y)\}_B$ , a constant  $\varepsilon > 0$ , and a constant  $C > 0$  such that for all balls  $B := B(c, r)$  we have*

$$(1.5) \quad \left\| \left( \frac{|x - c|}{r} \right)^{n(\frac{1}{p} - \frac{1}{s}) + \frac{\varepsilon}{s}} \|K(x, y) - P_{K,B}(x, y)\|_{L^q(B, dy/|B|)} \right\|_{L^s([B(c, 2r)]^c, dx)} \leq Cr^{n(s-1-1)},$$

where  $P_{K,B}(x, y) \equiv 0$  if  $r \geq 1$ ;